

**EDO STATE COLLEGE OF AGRICULTURE AND NATURAL
RESOURCES, IGUORIAKHI, NIGERIA**

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

COURSE CODE: AGT 124

COURSE TITLE: PRINCIPLES OF BEE KEEPING

CREDIT LOAD: 2 UNITS

AGT 124: BEE KEEPING

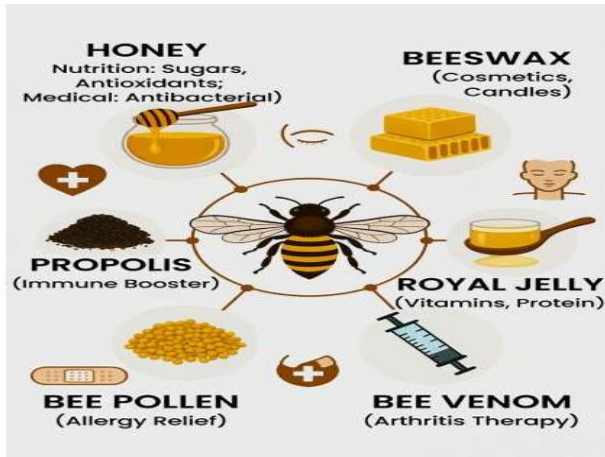


Goal: This course is designed to provide the student with the skill and knowledge required for bee keeping

General objectives: On completion of this course the students should be able to:-

- 1.0 Recognize the benefits and understand the history of beekeeping.
- 2.0 Know the conditions and requirements for successful apiary establishment and management
- 3.0 Recognize the equipments and basics of beekeeping processes.
- 4.0 Identify the activities and manipulation of bee in spring, summer, autumn and winter.
- 5.0 Understand the basics and tools of honeybee queen rearing.
- 6.0 Discuss the harvesting and processing of honey
- 7.0 Recognize the enemies of honey bees
- 8.0 Pesticide use & poisoning risks

UNIT 1: INTRODUCTION TO BEEKEEPING & HONEYBEE PRODUCTS



1.1 Different Products of Honeybee and Their Nutritive & Medicinal Values

Honeybees produce several useful products collectively called bee products. These are valuable for food, medicine, cosmetics, agriculture, and industry.

(A) Honey

Description: Sweet, viscous liquid produced by bees from flower nectar after enzymatic action and dehydration in the hive.

Nutritive Value:

- Carbohydrates: glucose and fructose – main energy source
- Vitamins: B-complex, Vitamin C
- Minerals: calcium, iron, potassium, magnesium
- Amino acids, enzymes, antioxidants

Medicinal Value/Uses:

- Treatment of cough, cold, and sore throat
- Healing wounds and burns :antibacterial properties
- Boosting immunity
- Improving digestion
- Providing instant energy
- Natural preservative

(B) Beeswax

Description: Produced by worker bees from 8 wax glands on the underside of the abdomen. Used to build comb.

Uses: Candle production, shoe polish, cosmetics like lip balm and creams, pharmaceutical preparations, lubricants, soap making, food wraps, wood polish.

Medicinal Value: Used in ointments for skin protection, burns treatment, and moisturizing creams.

(C) Propolis (Bee Glue)

Description: Sticky resinous substance collected from tree buds and bark, mixed with beeswax and enzymes. Used to seal cracks and protect the hive.

Uses: Hive protection from bacteria and fungi, medicine production.

Medicinal Value: Acts as antibacterial, anti-inflammatory, antifungal agent, and immune booster. Used in throat sprays and mouthwash.

(D) Royal Jelly

Description: Milky secretion from hypopharyngeal and mandibular glands of worker bees. Food for all larvae for 3 days and sole food for queen larvae/adult queen.

Nutritional Value: Rich in proteins, B-vitamins especially B5 & B6, hormones, amino acids, fatty acids.

Medicinal Value: Claimed for fertility improvement, memory enhancement, anti-aging treatments, body strength improvement. Used as dietary supplement.

(E) Bee Venom

Description: Poison produced in venom sac of worker bees, injected during stinging.

Medicinal Value: Used in apitherapy for treatment of arthritis, rheumatism, joint pain, and some nervous disorders. Caution: Can cause severe allergic reactions.

(F) Pollen / Bee Bread

Description: Pollen collected from flowers, mixed with nectar/enzymes, and packed in comb cells. Fermented pollen is called “bee bread”.

Nutritional Value: Contains 20-35% protein, vitamins, minerals, enzymes. Bee bread is more digestible due to fermentation.

Uses: Food supplement, animal feed enrichment, medicine production, probiotic for gut health.

1.2 Importance of Honeybees in Crop Pollination

Pollination is the transfer of pollen grains from the male part, anther, to the female part, stigma, of a flower, leading to fertilization.

Role of Honeybees in Pollination:

1. Transfer pollen between flowers as they forage for nectar and pollen.
2. Their hairy bodies trap pollen grains efficiently.
3. They show “flower constancy” : visit same species in one trip, improving pollination efficiency.

Importance to Agriculture:

1. Increase crop production and yield – 20-50% increase in many crops
2. Improve fruit quality, size, and shape
3. Increase seed formation and viability
4. Promote cross-pollination in self-sterile plants
5. Maintain biodiversity and ecosystem stability

Crops Pollinated by Bees in Nigeria: Cocoa, citrus, mango, cashew, oil palm, watermelon, vegetables like tomato and pepper, cowpea, melon, sesame, soybean, cotton.

Economic Note: Globally, pollination services by bees are valued higher than all hive products combined. Over 1/3 of food crops depend on insect pollinators.



1.3 Commercial Activity and Its Importance for National Income

Meaning of Commercial Beekeeping: Large-scale honey production and related activities using improved hives and management for sale and export, not just subsistence.

Commercial Activities in Beekeeping:

1. Honey production and processing
2. Beeswax processing
3. Pollen harvesting
4. Queen bee rearing
5. Hive construction and sales
6. Bee colony sales
7. Pollination service – renting colonies to farmers
8. Honey packaging and branding
9. Bee feed production

Importance to National Income:

1. Generates employment for youth and women in rural areas
2. Provides export earnings – Nigeria currently imports >80% of honey consumed
3. Supports rural development and poverty reduction
4. Improves farmers' income through increased crop yields via pollination
5. Supplies raw materials to food, pharmaceutical, and cosmetic industries
6. Diversifies agricultural GDP

1.4 List of Beekeeping Projects

Several projects promote beekeeping in Nigeria and these includes

1. Bees Abroad Pollinate Progress – trains smallholder farmers and youth.
2. UNDP Food Security Project (Benue State) – supplied kits and modern training to over 2,000 farmers, mostly women and youth since 2020.
3. Nigerian Montane Forest Project (Ngel Nyaki) – trains communities in improved hive designs for conservation and income.
4. Youths for Apiculture Initiative (YFAI) – partners with Federal Ministry of Livestock Development for youth engagement and World Bee Day events.
5. GlobalGiving Rural Beekeepers Enterprise (Kaduna) – aims to establish 4,000 rural beekeepers as an enterprise.

6. Slasita Farms training programmes – offers modern beekeeping and honey production workshops nationwide.
7. Bees for Self-Reliance Initiative – focuses on sustainable livelihoods through bees.



These projects provide training, equipment and market access, empowering communities and boosting honey production.

Students/beekeepers can undertake these as enterprises:

1. Honey production project
2. Beeswax processing project
3. Hive construction project
4. Queen rearing project
5. Bee colony multiplication project
6. Pollination service project
7. Honey packaging and branding project
8. Bee feed production project
9. Community apiary – group-owned 20-50 hives
10. Beekeeping training and consultancy

1.5 Laws Protecting Beekeeping Processes

Beekeeping laws protect bees, beekeepers, consumers, and the public. Nigeria has no standalone “Beekeeping Act” yet, but several laws apply:

1. Environmental Protection Laws: Prevent bush burning near apiaries and excessive pesticide spraying. Many states prohibit uncontrolled bush burning which destroys wild colonies and forage.

2. Agricultural Regulations: Control movement of bee colonies across states to prevent disease spread. Regulate importation of bees/queens.

3. Food Safety Laws – NAFDAC/SON: Ensure pure honey production, hygienic packaging, and labeling. Combat adulteration.

4. Forestry Laws: Protect bee forage plants and natural habitats in forest reserves. Some states require permits to site apiaries in reserves.

5. Public Safety: Most state bye-laws require apiaries to be sited 100-200m from dwellings, schools, and markets to prevent stinging incidents.

Best Practice: Beekeepers should register with Agricultural Development Programme ADP and Beekeepers Association of Nigeria for extension support and legal backing.

1.6 Past, Present and Future Situation of Beekeeping

Past Situation:

- Traditional beekeeping methods: clay pots, tree hollows, log hives, raffia/bamboo hives
- Characteristics: Low honey production 3-5 kg/hive/year, poor hive management, honey hunting often killed colonies, limited knowledge

Present Situation:

- Modern beekeeping using improved hives: Langstroth, Kenya Top Bar Hive
- Use of protective clothing, smokers, modern harvesting equipment
- Characteristics: Higher production 10-20 kg/hive/year, better training via ADPs/NGOs/universities, increased commercialization, rising market demand but adulteration is a problem

Future Situation:

- Expectations: Mechanized honey processing and extraction, increased export opportunities if quality standards improve, youth participation, more government support programs, improved bee breeding techniques, use of ICT for hive monitoring and e-commerce, growth of pollination services as a business

Threats: Climate change, deforestation, pesticide misuse, urbanization reducing forage.

1.7 Beekeeping in Nigerian Regions

Beekeeping varies across Nigeria's ecological zones. Northern regions (North-East, North-West, NorthCentral) practise mostly traditional methods using log, woven-straw, gourd and earthen-pot hives; savanna vegetation supports high commitment but faces security and deforestation issues. South-West (Lagos, Oyo, Ondo, Ogun, Osun, Ekiti) adopts modern Langstroth and top-bar hives rapidly due to rainforest forage; training programmes increase yields. South-East and South-South feature mixed systems integrated with cocoa and vegetable farms; humid climate aids colony growth but pests are common. Overall, northern states lead in traditional scale while southern states excel in commercial modernisation. Regional differences stem from vegetation, rainfall and culture. National projects bridge gaps through technology transfer, creating unified growth potential nationwide.

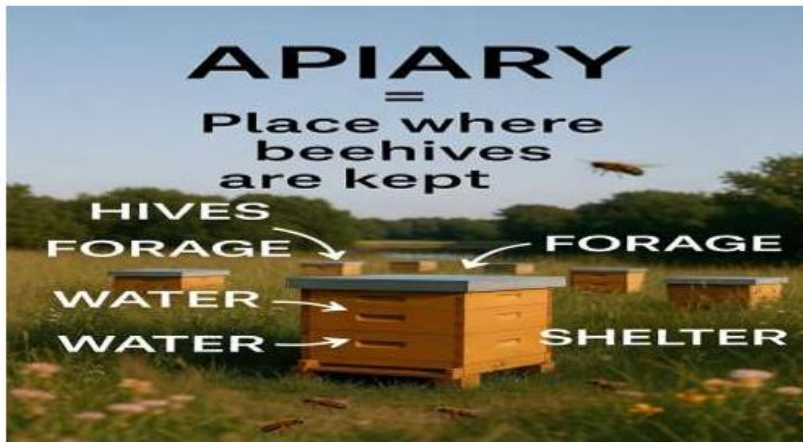


UNIT 2: APIARY ESTABLISHMENT

2.1 Identify the Apiary

Meaning of Apiary: An apiary is a designated location where managed colonies of honeybees are kept in hives for production and pollination. Also called **bee yard**.

It serves as the operational base for beekeepers. Key features include multiple hives arranged in rows, access paths, water sources, windbreaks and abundant flowering plants within 3 km radius. Apiaries must be sited away from human dwellings, livestock and busy roads for safety. Proper identification involves noting hive types, colony strength and management records. In Nigeria, apiaries range from backyard (small-scale) to commercial farms. Regular inspection for health and productivity defines a functional apiary. Understanding apiary layout ensures efficient management and higher honey yields.



Importance of Apiary:

1. Houses bee colonies in an organized manner
2. Enables honey and other products production
3. Protects bees from pests and bad weather
4. Allows systematic hive monitoring, inspection, and record keeping

2.2 Types of Apiaries

Apiaries are classified by hive design and management:

(A) Backyard Apiary

Located near the beekeeper's house. 1-5 hives.

Advantages: Easy monitoring, low cost, good for beginners.

Disadvantage: Risk to neighbors if not well managed.

(B) Commercial Apiary

Large-scale hive placement, 50+ hives, run as a business.

Advantages: High honey production, significant income generation.

Requires: Full-time management, capital, record keeping.

(C) Temporary Apiary

Used seasonally during flowering season, then removed.

(D) Migratory Apiary

Beehives moved from place to place to follow flowering crops and nectar flows.

Purpose: Maximize honey yield by chasing multiple flows per year. Higher yield but needs transport and labor.

(E) Fixed Apiary

Hives remain in one location year-round. Suitable where forage is available most months. Lower cost than migratory.

(F) Research/Observation Apiary

For training and demonstration. Often uses observation hives with glass sides.

2.3 Conditions and Requirements for Commercial Apiary Establishment and Management

A. Site Selection Requirements

A good apiary site must have these factors:

1. **Adequate Forage:** Presence of abundant flowering plants, trees, and crops within 2-3 km radius. Bees fly up to 5 km but are most efficient within 3 km.
2. **Water Supply:** Clean, permanent water must be available within 500m. Bees use water to cool hive and dilute honey. Provide in shallow bowls with floaters to prevent drowning.
3. **Security:** Protection from theft, animals like lizards and ants, and vandalism. Fence the area if possible.
4. **Accessibility:** Easy road access for inspection, harvesting, and transportation. But avoid sites too close to highways – vibrations disturb bees.
5. **Shade:** Protects bees from excessive sunlight and heat stress. Light morning sun is good to help bees start work early. Use trees for natural shade.
6. **Wind Protection:** Use trees, fences, or natural landforms as windbreaks. Strong wind topples hives and chills brood.
7. **Distance from Residential Areas:** Site 100-200m away from houses, schools, markets to prevent bee attack incidents. Face hive entrances away from footpaths.
8. **Good Drainage:** Gentle slope, no waterlogging. Dampness causes disease. Use hive stands to keep hives 30-45 cm above ground.
9. **Pesticide Safety:** Far from farms with heavy pesticide use. Pesticides are the #1 killer of bees. Liaise with nearby farmers on spray schedules.

UNIT 3: EQUIPMENT AND BASICS OF BEEKEEPING

DISTINGUISHING THE VARIOUS TYPES OF BEEHIVES AND THEIR COMPONENTS

1. Langstroth Hive

- Components:

- Outer Cover (protects from weather)
- Inner Cover (insulation and ventilation)
- Super (for honey storage)
- Queen Excluder (prevents queen from entering honey supers)
- Brood Chamber (for raising brood)
- Bottom Board (foundation of the hive)

- Measurements:

- Standard 10-frame: 50 cm (L) × 41 cm (W) × 24 cm (H)

2. Dadant Hive

- Similar to Langstroth but larger brood chamber.

- Components:

- Deeper brood boxes
- Same accessory components as Langstroth

- Measurements:

- 10-frame brood box: 50 cm (L) × 43.5 cm (W) × 31 cm (H)

3. Two-Queen Hive

- Components:

- Double brood chambers
- Dividing screen
- Multiple supers
- Dual entrances

- Purpose: Enhanced productivity via dual queen management.

4. Observation Hive

- Used for education and research

- Components:
 - Transparent side walls
 - Single or double frame
 - Small ventilation holes
- Measurements:
 - Vary depending on frame; typically, compact and portable

TYPES OF WAX FOUNDATION

In beekeeping, a wax foundation is a thin sheet of beeswax or plastic (coated with beeswax) that is inserted into a wooden or plastic frame and placed inside a beehive. It serves as a starter or guide for bees to build their honeycomb in an orderly and efficient manner.

Purpose and Benefits of Wax Foundation:

1. Guides Comb Building: Bees naturally build hexagonal cells, but the foundation gives them a structured guide, helping ensure straight and uniform combs.
2. Increases Hive Efficiency: Saves bees time and energy, as they don't have to start comb construction from scratch.
3. Strengthens the Comb: Helps support the comb, especially important during honey extraction when centrifugal force is used.
4. Prevents Cross-Combing: Keeps bees from building comb in random directions, which can make hive inspections difficult.

Types of Wax Foundation:

1. Pure Beeswax Foundation:
 - Made entirely from beeswax.
 - Preferred in natural or organic beekeeping.
 - Fragile and requires careful handling.
2. Plastic Foundation Coated with Beeswax:
 - More durable and easier to install.
 - Reusable but may be less accepted by bees at first.
3. Embossed Pattern:
 - The foundation is typically embossed with a hexagonal pattern to mimic natural honeycomb cell bases and encourage faster comb building.

Foundationless Beekeeping:

Some beekeepers choose to go foundationless, allowing bees to build comb naturally. This method is more natural but can be messier and harder to manage.

TOOLS FOR WIRING AND EMBEDDING

Wiring and embedding tools are essential when preparing wax foundation frames, especially for deep frames used in brood or honey supers. Wiring provides structural support to the wax foundation so it doesn't sag or break, and embedding ensures the wires are firmly pressed into the wax. Below are the key tools used:

Tools for Wiring Frames

1. Frame Wire

- Stainless steel or galvanized wire used to reinforce the wax foundation.
- Threaded horizontally or vertically across the wooden frame.

2. Wire Crimper (or Spur Embedder)

- Used to tighten and crimp the wire once threaded through the frame.
- Helps to prevent sagging by adding tension.

3. Eyelets and Eyelet Punch

- Small metal inserts placed in holes drilled into the sidebars of the frame.
- Prevent the wire from cutting into the wood and reduce friction.
- The eyelet punch is used to press eyelets into the holes.

4. Wire Tensioner

- A tool that applies tension to the wire while it's being installed, ensuring it stays taut.

5. Frame Jig or Wiring Board (optional)

- A stand or platform to hold the frame steady during wiring.
- Speeds up the process and ensures neat work.

Tools for Embedding Wire into Wax

1. Electric Embedder

- A device that uses low-voltage electricity to heat the wire.
- The warm wire melts into the wax foundation slightly, holding it in place.
- Commonly powered by a 12V battery or transformer.

2. Manual Spur Embedder (or Roller Embedder)

- A wheel-like hand tool with a grooved wheel that is rolled over the wire, pressing it into the wax.
- Ideal for areas without electricity.

3. Battery or Transformer

- Supplies 12 volts of DC power to the electric embedder.
- Must be used carefully to avoid overheating and damaging the wax.

PRODUCTION OF SECTION COMB HONEY

- Tools Needed:

- Section boxes (wooden or plastic)
- Foundation sheets (thin super)
- Comb cutters

- Process:

1. Fit foundation into section boxes.
2. Place in special super.
3. Allow bees to build comb and fill with honey.
4. Remove, cut, and package comb sections.

OTHER TYPES OF FOUNDATION AND COMBS

- Natural Comb: No foundation, bees build freely.
- Wired Wax Foundation: Reinforced for extraction.
- Perforated Foundation: Enhances airflow.

EQUIPMENT FOR INSPECTING COLONIES

Colony inspection is a routine but crucial activity to assess the health, productivity, and status of the hive. To do this safely and effectively, beekeepers use specific equipment for inspecting colonies.

Here's a comprehensive list of the essential tools and equipment:

1. Protective Gear

Used to protect the beekeeper from stings during inspection.

- Bee Suit or Jacket Full-body suit or upper-body jacket made of sting-proof material.
- Veil Protects the face and neck from bee stings.
- Gloves Leather or rubber gloves to protect hands; some experienced beekeepers may work gloveless for better dexterity.
- Boots Closed shoes or boots with pants tucked in to prevent bee entry.

2. Smoker

- A smoker produces cool smoke that calms the bees by masking alarm pheromones and encouraging them to consume honey, which makes them less aggressive.
- Fueled with materials like dry leaves, pine needles, cardboard, or wood shavings.

3. Hive Tool

- A flat, crowbar-like metal tool used to:
 - o Pry apart hive boxes and frames stuck with propolis.
 - o Scrape off wax and propolis.
 - o Lift or move frames for inspection.

Types: Standard hive tool, J-hook hive tool, Italian-style hive tool.

4. Bee Brush

- A soft-bristled brush used to gently remove bees from frames without harming them.
- Useful during inspections or when harvesting honey.

5. Frame Grip (Optional)

- A clamping tool that allows easy lifting of frames without fingers entering between tight frames.
- Helps prevent crushing bees.

6. Inspection/Record Book or App

- For documenting observations such as:
 - Queen presence
 - Brood pattern
 - Honey stores
 - Pest and disease symptoms
 - Colony strength

7. Sugar Spray or Water Spray Bottle (Alternative to Smoke)

- Used to gently calm bees when smoke is unavailable or not preferred.
- Sugar water distracts bees by encouraging feeding.

8. Varroa Mite Testing Kit (for health checks)

- Includes materials for checking Varroa mite infestation, such as:
 - Alcohol wash or powdered sugar jar

- Mite counter jar

9. Queen Marking Kit (Optional)

- Used to safely catch and mark the queen with a colored dot for easy identification during future inspections.

10. Flashlight or Headlamp (Optional)

- Useful for inspecting hives during low light or checking for eggs and larvae in dark comb.

INSPECTING THE COLONY

Times and Periods for Inspection

- Early Spring: Post-winter survival check
- Mid-Spring: Queen activity and brood development
- Pre-Nectar Flow: Colony strength and swarming tendency
- Post-Nectar Flow: Honey harvesting readiness
- Late Summer: Pest and disease check
- Autumn: Food reserves check
- Winter: Minimal disturbance, occasional external checks

Correct Opening and Inspection Procedure

1. Approach calmly
2. Use smoker sparingly
3. Open hive slowly and inspect top to bottom
4. Check for queen, brood, honey, pollen, diseases
5. Close gently

OBJECTIVES AND CYCLE OF INSPECTION

Objectives

- Monitor queen performance
- Check brood pattern and bee population
- Detect pests and diseases
- Evaluate food reserves

- Prevent swarming

Cycle of inspection

1. Cyclic Inspection

- Regular intervals (every 7-10 days during active season)
- Adjust frequency based on seasonal activity

Seasonal Inspection Details

- Spring: Feed if needed, replace old combs
- Summer: Add supers, control swarms
- Autumn: Reduce hive size, feed syrup
- Winter: Ensure ventilation, insulation

ARRANGEMENT OF COLONIES IN SPRING

- Reorganize frames (brood at center, food at sides)
- Clean and repair equipment
- Equalize colony strength if necessary

INSPECTION DURING NECTAR FLOW SEASON

- Add supers (boxes or hive bodies that are placed above the brood chamber in a beehive. The main purpose of supers is to provide space for bees to store surplus honey) as needed
- Ensure enough space for incoming nectar
- Remove capped honey

INSPECTION AFTER NECTAR FLOW

- Assess harvested honey
- Begin mite treatment
- Feed if stores are low

SUMMER INSPECTION

- Check for overheating, adequate ventilation

- Monitor queen and brood
- Observe for robbing

AUTUMN INSPECTION

- Confirm winter stores (15–20 kg of honey minimum)
- Combine weak colonies
- Final mite treatment

WINTER INSPECTION

- Visual checks
- Confirm hive entrance is clear
- Observe hive sounds and activity

SIGNIFICANCE OF A LOG BOOK OF INSPECTION

- Records queen performance and health trends
- Helps track feeding and treatments
- Aids in swarming and disease management
- Supports decision-making in colony management

for better yields/health. Essential professional tool in beekeeping.

UNIT 4: Identify the activities and manipulation of bee in spring, summer, autumn and winter.

4.0 Identifying Activities and Manipulation of Bees in Spring, Summer, Autumn, and Winter

Honey bee colonies follow seasonal cycles tied to temperature, forage availability, and survival needs.

- Winter — Bees form a tight cluster for warmth; minimal activity; consume stored honey.
- Spring — Rapid brood rearing and population buildup; foraging resumes.
- Summer — Peak foraging and honey storage; high activity but risk of swarming.
- Autumn — Colony reduces brood; stores honey for winter; prepares for cold.

4.1 Activities of Bees in Winter

In cold weather, bees stop flying and cluster to generate heat by vibrating muscles. The colony shrinks; the queen stops laying; bees eat stored honey/pollen. Survival depends on food stores and insulation.

4.2 Clustering

Clustering is the winter survival strategy. Bees form a ball around the queen: outer bees (mantle) insulate, inner bees generate heat (core ~92°F/33°C even in freezing temps). Bees rotate positions to share warmth. The cluster tightens in cold and loosens in milder weather.



Illustration of a winter bee cluster inside a hive, showing tight ball formation for heat conservation.

4.3 & 4.4 Wintering and Feeding / Requirements

Wintering requirements: Strong colony (enough bees), young queen, 30–60+ lbs honey stores, good ventilation (to avoid moisture), wind protection, and insulation in cold climates. Monitor stores by hefting hives.

Feeding if short: Use fondant/sugar cakes (dry, no drowning risk) or 2:1 syrup in autumn. Avoid disturbing cluster.

Feeders

- Entrance feeders (jar at entrance)
- Top feeders (hive-top containers)
- Baggie feeders (slit plastic bags)
- Frame feeders (inside hive)
- Candy boards (solid sugar)

4.8 Activities of Bees in Spring

Days lengthen → queen resumes laying; brood expands rapidly; pollen/nectar foraging surges; population booms; drones appear; surplus stores build. Bees become active and less defensive.

4.9 Swarming

Swarming is natural reproduction: Colony rears new queens; old queen leaves with ~50–75% workers/drones to form new colony. Causes: Overcrowding, abundant food, old queen pheromones decline. Signs: Queen cells (peanut-shaped).

Prevention: Add space (supers), reverse boxes, young queens



Photo of a honey bee swarm cluster on a branch, typical post-swarm behavior.

4.10 Absconding (Migration)

Absconding differs from swarming: Entire colony (queen + all bees) abandons hive due to stress (e.g., severe pests like Varroa, starvation, water issues, disease). No queen cells left; no reproduction. More common in stressed or Africanized bees.

4.11 Robbing: Reasons, Detriment, Protection/Control

Robbing — Strong bees steal honey from weak colonies during nectar dearth (autumn/late summer). Reasons: Food scarcity; weak guards in target hive.

Detriment — Spreads diseases (e.g., foulbrood); kills bees in fights; destroys weak colonies.

Control — Reduce entrances; feed colonies simultaneously; avoid spills; use robbing screens.

4.12 Laying Workers

Laying workers occur in queenless colonies (no queen/brood pheromones suppress ovaries). Workers lay unfertilized (drone) eggs → scattered, multiple eggs/cell; high drone population.

Detriment — Colony fails (no workers reared long-term).

Control — Add open brood/queen cell to suppress; unite with queen-right colony; shake bees out (workers return home, laying ones lost).

4.13 Colony Dividing or Artificial Swarming

Artificial swarming mimics natural swarming to prevent loss/control growth. Methods (e.g., Pagden):

- Find queen → move her + flying bees to new hive location.
- Leave brood/queen cells in original spot → bees raise new queen.
- Flying bees return to new site → split colony.

4.14 Activities of Bees in Summer

Peak foraging; maximum population; honey surplus stored; drones abundant; high defense; risk of swarming/overheating.

4.15 Protecting Bees in Summer

Provide shade/ventilation; ensure water; add supers for space; monitor pests; prevent robbing/swarming.

4.16 Importance of Saving Water for Bees

Bees need water for cooling hive (fanning evaporative cooling), diluting honey, brood rearing. Provide shallow sources with landing spots (e.g., pebbles in trays) to prevent drowning.

UNIT 5: Understand the basics and tool of honey bee queen rearing

Tools used in the rearing of honeybee queen

Rearing honeybee queens involves specialized tools and techniques to ensure the production of high-quality queens.

Tools:

1. **Queen cups or cells:** Small, cup-shaped structures where queen larvae are raised.
2. **Grafting tools:** Specialized tools used to transfer larvae from worker cells to queen cups.
3. **Queen excluders:** Devices that allow worker bees to pass through but keep the queen from leaving the hive.
4. **Beehives:** Specifically designed hives for queen rearing, such as nucleus colonies (nucs) or queen starter colonies.

Methods:

1. **Grafting:** Transferring larvae from worker cells to queen cups to initiate queen development.
2. **Queen cell construction:** Building queen cups or cells to accommodate the developing queens.
3. **Larval selection:** Selecting suitable larvae for grafting, typically 1-2 days old.
4. **Nucleus colonies:** Using small colonies (nucs) to raise and mate queens.
5. **Mating stations:** Isolated areas where queens are allowed to mate with selected drones.

Types of queen production

A. Natural queen production

Natural queen production in honeybees occurs in several situations:

1. **Swarm preparation:** When a colony prepares to swarm, it raises new queens to replace the existing queen and to populate the new swarm.
2. **Queen supersedure:** When a colony's queen is aging, injured, or failing, the colony may raise a new queen to replace her.
3. **Queen loss:** If a colony loses its queen due to disease, predation, or other factors, the colony may raise a new queen from existing brood.
4. **Colony fission:** In some cases, a colony may divide, and a new queen is raised to lead the new colony.

In these situations, the bees typically select larvae and feed them royal jelly to induce queen development. The resulting queens are genetically diverse and can exhibit unique characteristics.

B. Controlled queen rearing

Controlled queen rearing involves manipulating honeybee colonies to produce high-quality queens with desired characteristics.

Methods:

1. **Grafting:** Transferring larvae from a breeder queen's colony to queen cups, where they're raised into queens.
2. **Larval selection:** Choosing larvae from high-performing colonies or breeder queens with desirable traits.
3. **Queen cell construction:** Building queen cells or cups to accommodate developing queens.
4. **Nucleus colonies (nucs):** Using small colonies to raise and mate queens.

Controlled Conditions:

1. **Selected breeder queens:** Choosing queens with desirable traits, such as high honey production or disease resistance.
2. **Controlled mating:** Managing drone populations to ensure mating with selected drones.
3. **Environmental control:** Regulating temperature, humidity, and nutrition to support queen development.

Benefits:

1. **Improved genetics:** Selecting for desirable traits enhances colony performance.
2. **Increased queen quality:** Controlled conditions promote healthy queen development.
3. **Predictable results:** Controlled queen rearing allows for more consistent outcomes.

By controlling queen rearing, beekeepers can:

1. **Improve colony performance**
2. **Enhance disease resistance**
3. **Increase honey production**
4. **Develop specific traits**

This method enables beekeepers to produce high-quality queens that meet specific needs and improve overall colony health.



UNIT 6: HARVESTING AND PROCESSING OF HONEY

6.1 Honey harvesting is the process of collecting ripe honey from bee colonies, while processing ensures it is clean, safe, and marketable without destroying its natural quality.



6.2 Step-by-Step Honey Harvesting Process

Step 1: Identify Mature Honey

Honey is ready when:

At least 75–80% of the comb is capped with wax

Moisture content is low (~18%)

Unripe honey ferments easily → must be avoided



Figure 59. A fully capped frame of honey ready to be removed from a colony and extracted. (Courtesy Dewey Caron)

Step 2: Prepare Equipment

Before harvesting, ensure availability of:

Bee suit (protective clothing)

Smoker

Hive tool

Bee brush

Honey extractor

Buckets/containers

Step 3: Calm the Bees

Use a smoker to release cool smoke into the hive

Smoke masks alarm pheromones and reduces aggression

Step 4: Remove Honey Frames

Open hive gently using hive tool

Remove frames containing capped honey

Shake or brush off bees carefully

Step 5: Transport to Extraction Area

Keep frames in covered containers

Prevent contamination and robbing by bees

6.3 Honey Processing Steps

Step 6: Uncapping the Honey

Use an uncapping knife or fork

Remove wax caps sealing the honeycomb cells



Figure 62. Wax cappings are removed from a honey comb with a hot knife. (Courtesy Dewey Caron)

Step 7: Extraction

Place frames in a honey extractor

Spin to remove honey using centrifugal force

Step 8: Filtration

Filter honey through mesh or cloth

Removes:

Wax particles

Dead bees

Debris

Step 9: Settling

Allow honey to stand for 24–48 hours

Air bubbles and impurities rise to the top

Step 10: Packaging and Storage

Store in clean, airtight containers

Keep in a cool, dry place

Avoid moisture contamination

EQUIPMENT USED IN HONEY EXTRACTION

6.4 Description of Equipment

1. Bee Suit (Protective Clothing)

Full-body covering including veil

Prevents bee stings

2. Smoker

Produces smoke from fuel (wood shavings, dry leaves)

Calms bees during hive inspection

3. Hive Tool

Metal tool used for:

Opening hive

Scraping wax and propolis

4. Bee Brush

Soft brush used to gently remove bees from frames

5. Uncapping Knife/Fork

Used to remove wax cappings from honeycomb

6. Honey Extractor

Manual or electric device

Spins frames to extract honey

7. Strainers/Filters

Fine mesh sieves

Used for cleaning honey

8. Storage Tanks/Buckets

Food-grade containers

Used for settling and storing honey

6.5 Correct Use of Honey Extraction Equipment

1. Bee Suit

Wear properly and ensure no openings

Prevents stings and panic

2. Smoker

Use cool smoke (not hot)

Do not over-smoke → can affect honey flavor

3. Hive Tool

Use gently to avoid damaging frames

Clean after use to prevent disease spread

4. Uncapping Knife

Use at correct temperature (warm knife works best)

Avoid cutting too deep into comb

5. Honey Extractor

Balance frames evenly

Spin gradually (slow → fast)

Clean thoroughly after use

6. Filters and Containers

Always sterilize before use

Avoid contamination with water or dirt

UNIT 7: ENEMIES OF HONEY BEES

Honey bees face threats from:

- Pests
- Diseases
- Predators
- Chemicals (pesticides)

7.1 Honey Bee Pests

1. Wax Moths

Destroy combs

Lay eggs in hive → larvae feed on wax

2. Hive Beetles

Damage honey and brood

Cause fermentation of honey

3. Ants

Invade hive and steal honey

4. Termites

Attack wooden hive structures

7.2 Honey Bee Diseases

1. American Foulbrood (AFB)

Bacterial disease

Symptoms:

Dead larvae with foul smell

Sticky brown brood

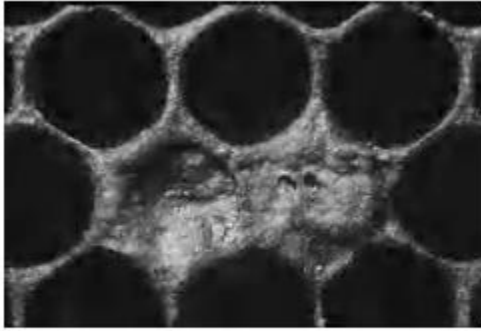


Figure 36. Brood cells from a colony with American foulbrood, showing sunken, punctured cappings.
(Courtesy Scott Camazine)

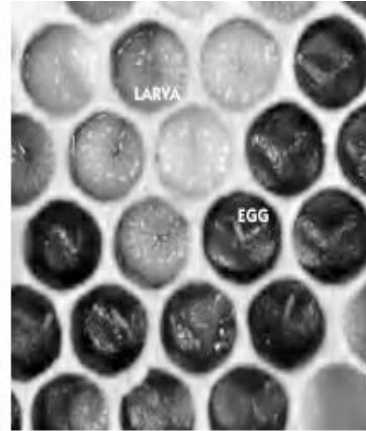


Figure 6. Cells with healthy worker larvae.
(Courtesy Dewey Caron)

2. European Foulbrood (EFB)

Affects young larvae

Larvae appear twisted and discolored

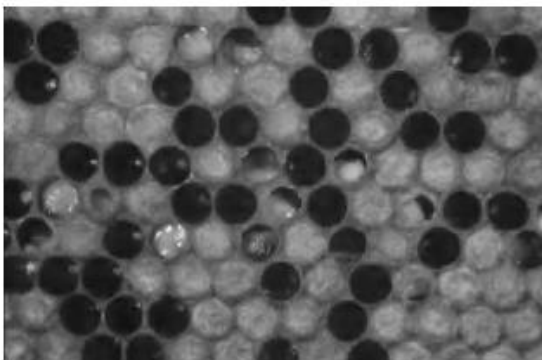


Figure 39. Larvae infected with European foulbrood.
(Courtesy Maryann Frazier)

3. Sac brood

This disease, caused by a virus, does not usually cause severe losses.

Both worker and drone larvae are affected. Pupae may be killed occasionally, whereas adult bees are immune to it. Death usually occurs after the cell is sealed and the larva has spun its cocoon. Dead brood is often scattered among healthy brood. The cappings over dead brood are first punctured and later removed by the bees. The larvae gradually change from pearly white to dull yellow or gray and finally to black. The head of the larva, the first part of the body to change color, becomes black. Larvae die in an upright position with raised heads (Figure 41).

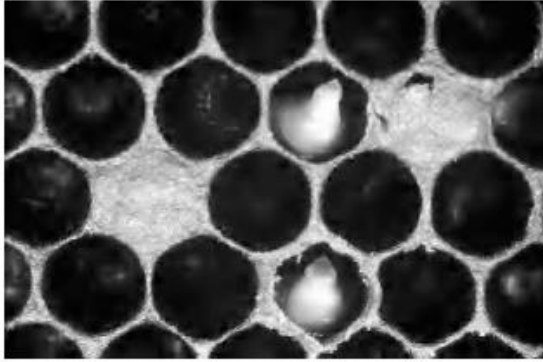


Figure 41. Dead larva infected by sacbrood with typical darkened, raised head. (Courtesy Scott Camazine)

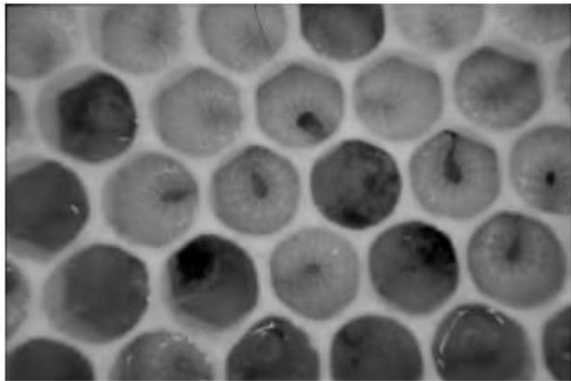


Figure 40. European foulbrood advanced; larva twisted in cell. (Courtesy Dewey Caron)



Figure 42. Larva infected with sacbrood that has been removed from the cell. Note the sac-like appearance of the larva when removed. (Courtesy Scott Camazine)

4. Nosema Disease

Caused by protozoa

Symptoms:

Diarrhea

Weak colony

4. Chalkbrood

Fungal disease

Larvae become hard and chalk-like

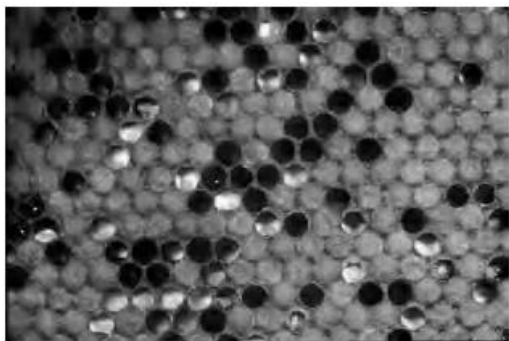


Figure 43. Worker larvae infected with chalkbrood. (Courtesy Scott Camazine)



Figure 44. Chalkbrood "mummies" removed by workers and dropped at hive entrance. (Courtesy Maryann Frazier)

7.3 Predators of Honey Bees

- Birds
- Lizards
- Frogs
- Wasp

UNIT 8 PESTICIDE USE & POISONING RISKS

8.1 Sources of Poisoning

- Agricultural pesticides
- Herbicides and fungicides
- Spraying during flowering period



8.2 Effects of Pesticides on Bees

- ✓ Immediate death
- ✓ Disorientation
- ✓ Reduced foraging ability
- ✓ Colony collapse

8.3 Signs of Poisoning

- ✓ Dead bees around hive entrance
- ✓ Weak or restless bees
- ✓ Reduced honey production

8.4 Prevention of Poisoning

For Farmers:

Avoid spraying during flowering

Spray in evening when bees are inactive

Use bee-friendly pesticides

For Beekeepers:

Locate apiary far from sprayed fields

Provide clean water sources

Close hive temporarily during spraying

8.5 Safe Use of Pesticides

Follow label instructions strictly

Use recommended dosage

Avoid mixing harmful chemicals